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Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.

2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.

3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.

4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.

5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.

6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.

7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

8. Alan V. Oppenheim, Alan S. Willsky, with S. Hamid Nawab, *Signals and Systems*, 3rd edition, Pearson, 2023.

9. Simon Haykin, Barry Van Veen, *Signals and Systems*, 2nd edition, Wiley, 2003.

10. Steven H. Strogatz, *Nonlinear Dynamics and Chaos: With Applications to Physics, Biology, Chemistry, and Engineering*, 2nd edition, Westview Press, 2018.

11. Norman S. Nise, *Control Systems Engineering*, 7th edition, Wiley, 2015.

EE1060: Differential Equations and Transform Techniques

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Topics covered :

1. Construction of Fundamental set of solutions

- We consider a homogeneous first-order system of differential equations given by:

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \tag{1}$$

  where $\mathbf{x}(t)$ is an n-dimensional vector of functions and $\mathbf{A}$ is a constant $n \times n$ matrix. The primary objective is to determine a *fundamental set of solutions*, comprising n linearly independent solutions that span the complete solution space of the system.

- From the study of nth-order linear differential equations, it is well known that exponential functions of the form $e^{\lambda t}$ frequently arise as solutions. This naturally motivates the use of exponential trial solutions when solving systems of first-order linear differential equations.

- To establish the connection formally, we observe that an nth-order scalar differential equation can be reformulated as a first-order system. This is done by introducing a state vector that captures the function and its first $n-1$ derivatives:

$$\mathbf{x}(t) = \begin{bmatrix} x(t) \\ \dot{x}(t) \\ \vdots \\ x^{(n-1)}(t) \end{bmatrix}. \tag{2}$$

  The original scalar differential equation is thus represented as:

$$\frac{d\mathbf{x}}{dt} = \mathbf{A}\mathbf{x}, \tag{3}$$

  where the matrix $\mathbf{A}$ is constructed using the coefficients of the original equation.

- Since $x(t) = e^{\lambda t}$ is a typical solution of the scalar nth-order equation, substituting this into the definition of the state vector yields:

$$\mathbf{x}(t) = \begin{bmatrix} e^{\lambda t} \\ \lambda e^{\lambda t} \\ \vdots \\ \lambda^{n-1} e^{\lambda t} \end{bmatrix} = \mathbf{v}e^{\lambda t}, \tag{4}$$

  where $\mathbf{v}$ is a constant vector whose components depend on λ. This motivates the trial solution:

$$\mathbf{x}(t) = \mathbf{v}e^{\lambda t}. \tag{5}$$

- Substituting the trial solution into the original system, we differentiate:

$$\frac{d}{dt}(\mathbf{v}e^{\lambda t}) = \lambda \mathbf{v}e^{\lambda t}, \tag{6}$$

  and compute the right-hand side:

$$\mathbf{A}\mathbf{x}(t) = \mathbf{A}(\mathbf{v}e^{\lambda t}) = \mathbf{A}\mathbf{v}e^{\lambda t}. \tag{7}$$

  Equating the two expressions and cancelling $e^{\lambda t}$, which is never zero, we obtain:

$$\mathbf{A}\mathbf{v} = \lambda \mathbf{v}. \tag{8}$$

  This is the standard eigenvalue equation. Thus, for every eigenvalue λ of $\mathbf{A}$ with corresponding eigenvector $\mathbf{v}$, the function $\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$ is a solution to the system.

- To construct the general solution, we aim to find n linearly independent solutions of the form $\mathbf{x}(t) = \mathbf{v}e^{\lambda t}$. This requires identifying n linearly independent eigenvectors of $\mathbf{A}$. The ability to do so depends on the algebraic and geometric multiplicities of the eigenvalues. Accordingly, we distinguish between three cases:

  1. **Distinct Real Eigenvalues:** If $\mathbf{A}$ has n real and distinct eigenvalues, then the corresponding eigenvectors are guaranteed to be linearly independent. Each eigenpair gives rise to an exponential solution, and their linear combination forms the general solution.
  2. **Complex Conjugate Eigenvalues:** When $\mathbf{A}$ has complex eigenvalues (which must occur in conjugate pairs if $\mathbf{A}$ has real entries), the corresponding eigenvectors are also complex. The resulting complex-valued solutions can be converted into real-valued functions using Euler's formula, typically involving sine and cosine terms.
  3. **Repeated Eigenvalues:** If some eigenvalues have multiplicity greater than one and the number of linearly independent eigenvectors is less than their multiplicity, then generalized eigenvectors must be used. These lead to additional solutions involving polynomials multiplied by exponentials.

- **Case: Distinct Real Eigenvalues.** Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be n distinct real eigenvalues of $\mathbf{A}$, with corresponding linearly independent eigenvectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$. Then, the solution corresponding to each eigenpair is:

$$\mathbf{x}_i(t) = \mathbf{v}_i e^{\lambda_i t}, \quad i = 1, 2, \ldots, n. \tag{9}$$

  The general solution is a linear combination:

$$\mathbf{x}(t) = \alpha_1 \mathbf{v}_1 e^{\lambda_1 t} + \alpha_2 \mathbf{v}_2 e^{\lambda_2 t} + \cdots + \alpha_n \mathbf{v}_n e^{\lambda_n t}, \tag{10}$$

  where $\alpha_1, \ldots, \alpha_n$ are constants determined by the initial condition $\mathbf{x}(0)$.

- These n linearly independent solutions form a fundamental set and fully describe the solution space when $\mathbf{A}$ has distinct real eigenvalues.